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REVIEW ARTICLE

Nanostructured material–based electrochemical sensing of oxidative DNA damage biomarkers 8-oxoguanine and 8-oxodeoxyguanosine: a comprehensive review

Ana-Maria Chiorcea-Paquim [1,2] · Ana Maria Oliveira-Brett [1,2]

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Abstract

Oxidative DNA damage plays an important role in the pathogenesis of various diseases. Among oxidative DNA lesions, 8-oxoguanine (8-oxoG) and its corresponding nucleotide 8-oxo-2′-deoxyguanosine (8-oxodG), the guanine and deoxyguanosine oxidation products, have gained much attention, being considered biomarkers for oxidative DNA damage. Both 8-oxoG and 8-oxodG are used to predict overall body oxidative stress levels, to estimate the risk, to detect, and to make prognosis related to treatment of cancer, degenerative, and other age-related diseases. The need for rapid, easy, and low-cost detection and quantification of 8-oxoG and 8-oxodG biomarkers of oxidative DNA damage in complex samples, urine, blood, and tissue, caused an increasing interest on electrochemical sensors based on modified electrodes, due to their high sensitivity and selectivity, low-cost, and easy miniaturization and automation. This review aims to provide a comprehensive and exhaustive overview of the fundamental principles concerning the electrochemical determination of the biomarkers 8-oxoG and 8-oxodG using nanostructured materials (NsM), such as carbon nanotubes, carbon nanofibers, graphene-related materials, gold nanomaterials, metal nanoparticles, polymers, nanocomposites, dendrimers, antibodies and aptamers, and modified electrochemical sensors.

Keywords 8-oxoguanine (8-oxoG) · 8-oxo-2′-deoxyguanosine (8-oxodG) · Oxidative DNA damage · Oxidative DNA damage biomarker · Electrochemical sensor · Nanostructured materials (NsM)

Introduction

DNA damage, such as single- and double-strand breaks, base oxidation, and intra- and inter-strand crosslinks, can be induced endogenously, during the DNA replication and by reactive oxygen species (ROS), and exogenously, by ionizing and UV radiation, chemical agents, tobacco smoking, industrial and diesel exhaust, xenobiotics, and microbial and viral pathogens (Scheme 1). In particular, oxidative DNA damage is associated with increased genomic mutagenesis and carcinogenesis and plays an important role in many autoimmune inflammatory diseases, neurodegenerative diseases, atherosclerosis, ischemia-reperfusion injury, stroke and chronic respiratory disease, cystic fibrosis, diabetes, and other age-related diseases.

The oxidative damage of genomic DNA results from the oxidation of the DNA bases, nucleosides, and nucleotides. Guanine (G) is the DNA base most susceptible to the action of ROS and presents the lowest oxidation potential [1]. Among the different G oxidation products [2–4], 8-oxoguanine (8-oxoG or 8-oxoGua), also known as 7,8-dihydro-8-oxoguanine, 8-dihydroguanine, or 8-hydroxyguanine (8-OHG) [5], and its corresponding nucleotide 8-oxo-2′-deoxyguanosine (8-oxodG), also known as 8-oxo-7,8-dihydro-2′-deoxyguanosine or 8-hydroxy-2′-deoxyguanosine(8-OHdG) (Scheme 2a), are the cause of the most predominant forms of oxidative DNA damage.

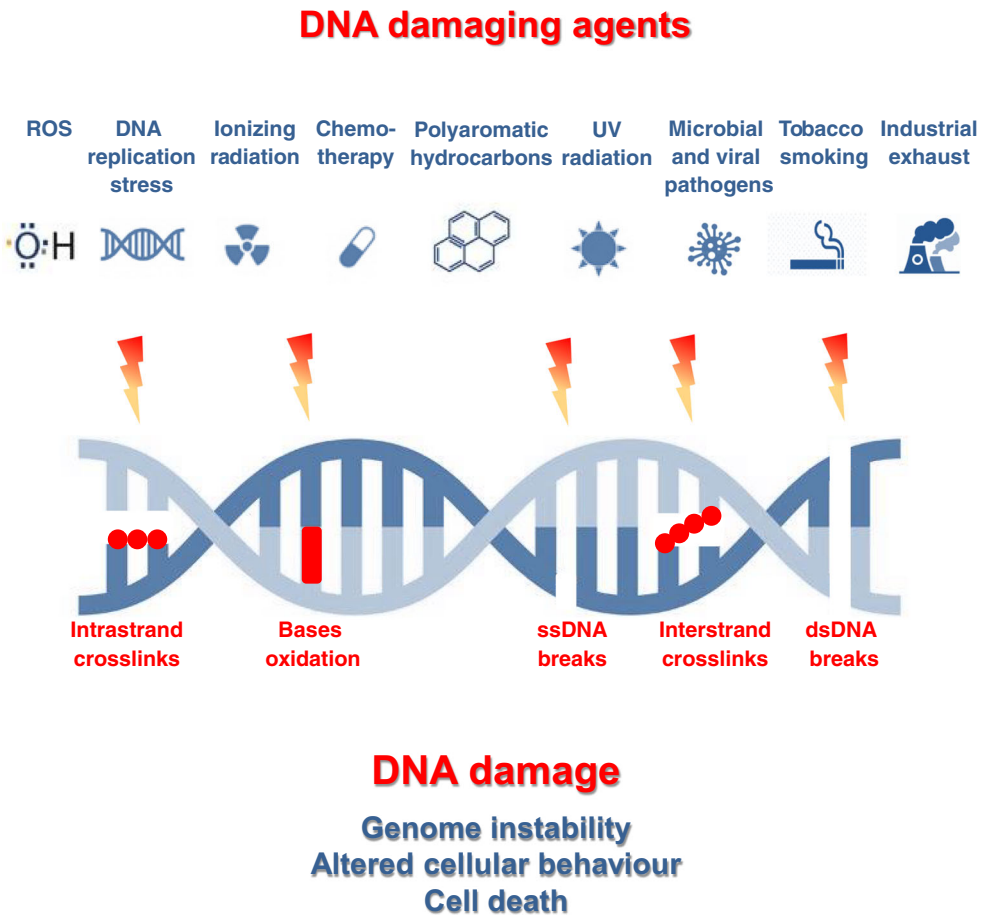
In fact, 8-oxoG, the product of oxidation of G at the C8 position (Scheme 3), presents several tautomeric forms in aqueous solution (Scheme 2b), the most stable being the 6,8-diketo I tautomer, with an oxo group at the C8 and an H atom at the N7 positions [6–9]. The incorporation of 8-oxoG into

✉ Ana Maria Oliveira-Brett
brett@ci.uc.pt

Ana-Maria Chiorcea-Paquim
https://orcid.org/0000-0002-0605-5147

[1] University of Coimbra, Centre for Mechanical Engineering, Materials and Processes (CEMMPRE), Department of Chemistry, 3004-535 Coimbra, Portugal

[2] Instituto Pedro Nunes (IPN), 3030-199 Coimbra, Portugal



Scheme 1 Schematic representation of DNA damage

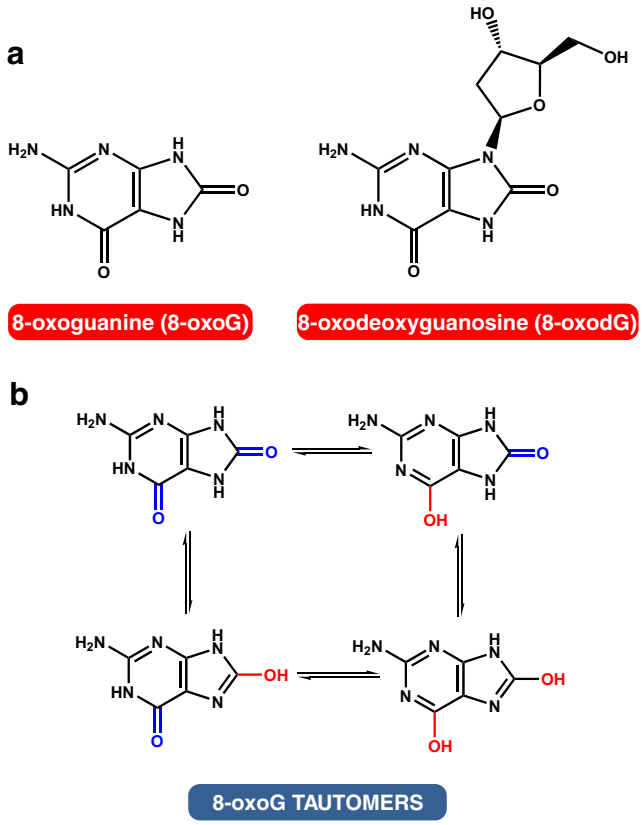
genomic DNA can occur either by direct oxidation of the G base in the DNA strand or by oxidation of the dGTP or GTP in the nucleotide pool that is further integrated into the DNA during replication or repair [10, 11].

The 8-oxoG lesion is highly mutagenic, leading to G:C-T:A and G:C-C:G transversions during DNA replication [2], and its repair in the DNA strand follows the base excision repair (BER) pathway initiated by the oxoguanine glycosylase 1 (OGG1) [8], which results in 8-oxoG urinary excretion [11]. Furthermore, the nucleotide pool is cleaned of 8-oxodGTP by the enzyme nucleoside diphosphate linked moiety X type motif 1 (NUDT1), which also results in 8-oxodG urinary excretion [11]. Therefore, the urinary excretion of 8-oxoG from the damaged nucleotides in the DNA, and of 8-oxodG from the cellular pools, reflects the balance of the rate of oxidative DNA damage in vivo and the efficacy of repair processes. The 8-oxoG and 8-oxodG levels in urine, and also in blood and tissue samples, have gained much attention, being considered biomarkers for oxidative DNA damage, and used to predict the overall body oxidative stress levels, to estimate the risk, to detect, and to make prognosis related to treatment of cancer, degenerative and other age-related diseases.

In the detection and quantification of 8-oxoG and 8-oxodG, high-performance liquid chromatography (HPLC) with electrochemical detection (HPLC-ECD) [12], HPLC tandem mass spectrometry (HPLC-MS), gas chromatography-mass spectrometry (GC-MS), capillary electrophoresis with electrochemical detection (CE-ECD), and enzyme-linked immunosorbent assay (ELISA), have been employed. Recently, electrochemical sensors based on modified electrodes gained great attention, due to their high sensitivity and selectivity, low-cost, and easily miniaturization and automation capabilities.

The most common method used for the detection of oxidative DNA damage biomarkers 8-oxoG and 8-oxodG was initially HPLC-ECD, which enabled the 8-oxodG determination in HeLa cells [13], mouse liver [13], rat kidney and urine [14], liver [15], heart [16], human leukocytes [17], blood mononuclear cells [18], lung [19], brain [20], breast [18], urine [21], and saliva [22]. Despite its high sensitivity, good selectivity, and low limit of detection (LOD), HPLC-ECD usually requires previous cleanup procedures and sample purification treatments. Moreover, during the HPLC-ECD measurements, problems may occur, due to the electrode poisoning, which result in current decrease and reproducibility problems.

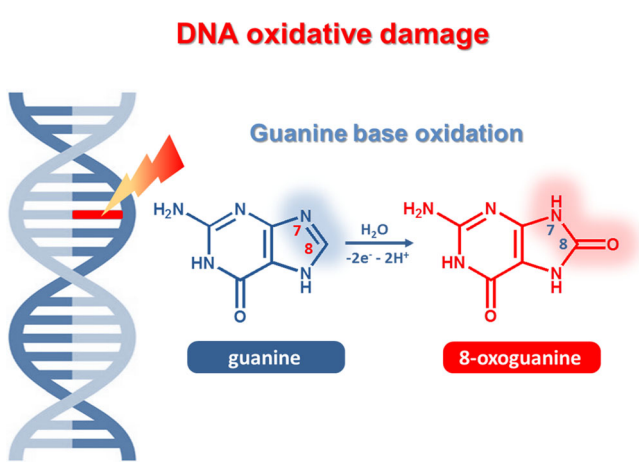
The need for rapid, easy, and low-cost detection and quantification of 8-oxoG and 8-oxodG biomarkers in complex urine, blood, and tissue samples, combined with the necessity of reducing the utilization of organic solvents due ecological



Scheme 2 **a** 8-Oxoguanine (8-oxoG) and 8-oxo-2′-deoxyguanosine (8-oxodG) chemical structures and **b** 8-oxoG tautomers

considerations, caused an increasing interest on voltammetric methods that are highly sensitive. Moreover, nanostructured materials (NsM) [23, 24], i.e. materials with a microstructure, the characteristic length scale of which is on the order of a few (typically 1–10) nanometres, have been increasingly used to modify the electrode surface of electrochemical sensors.

This review presents a comprehensive and exhaustive overview of 8-oxoG and 8-oxodG electroanalytical determination (Table 1) using electrochemical sensors modified by NsM,



Scheme 3 Schematic representation of guanine (G) oxidation to 8-oxoguanine (8-oxoG)

such as carbon nanotubes (CNT), carbon nanofibers, graphene related materials, gold nanomaterials, metal nanoparticles (NPs), polymers, nanocomposites, dendrimers, antibodies, and aptamers (Scheme 4).

Electrochemical sensing of 8-oxoG and 8-oxodG

Unmodified electrodes

The 8-oxoG oxidation at a bare glassy carbon electrode (GCE) was studied by cyclic voltammetry (CV), differential pulse voltammetry (DPV), and square wave voltammetry (SWV) [25]. The anodic peak, at E_p ~ + 0.58 V (vs. Ag/AgCl, 3 M KCl), in pH 3.6 (Fig. 1) corresponds to a reversible, pH-dependent process that occurs with the transfer of two electrons and two protons, in the pH range from 1.1 to 13.1. At higher potentials occurs the oxidation and polymerization of the 8-oxoG oxidation products (Fig. 1). The direct electroanalysis of 8-oxoG, at a GCE, using DPV, in pH = 4.5, in the linear range 6–28 × 10^{-6} M, with a LOD = 8 × 10^{-7} M, was performed [25].

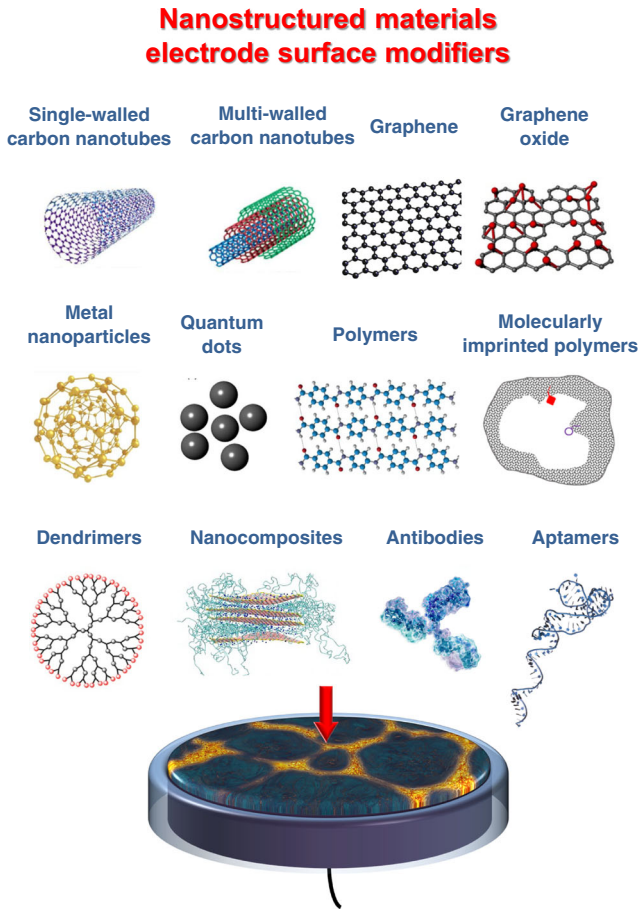
The 8-oxo-G oxidation at a boron-doped diamond electrode (BDDE), studied by DPV (Fig. 2), showed that the oxygen and hydrogen terminations, formed at the BDDE after surface pre-treatment, strongly influenced the 8-oxoG redox behaviour, peak potential and current [26]. An enhanced electrochemical response on the BDDE surface was obtained after cathodic electrochemical pre-treatment (Red-BDDE), when compared with the anodic electrochemical pre-treatment (Ox-BDDE) (Fig. 2). In pH = 4.5, the 8-oxoG oxidation at a Red-BDDE occurs at E_p = + 0.49 V (vs. Ag/AgCl, 3 M KCl), whereas at an Ox-BDDE is shifted to a higher potential, at E_p = + 0.62 V. Moreover, the OH• radicals formed at the Ox-BDDE surface interact with the 8-oxoG, leading to a decrease in the oxidation peak current. In pH = 7.0, the 8-oxoG oxidation at a Red-BDDE is shifted to a lower potential, at E_p = + 0.36 V, whereas at Ox-BDDE is not pH dependent, occurring at E_p = + 0.62 V [26]. The 8-oxoG electroanalytical determination at a Red-BDDE, in pH = 4.5, with a LOD = 1.0 × 10^{-6} M, and a limit of quantification LOQ = 3.2 × 10^{-6} M, was achieved [26].

The 8-oxodG oxidation at GCE and pyrolytic graphite electrode (PGE) also showed a two-electron, two-proton process, followed by irreversible chemical reactions, leading to the formation of diimine species, which undergo a series of chemical reactions to give rise to urea riboside, 2-amine-4,5,6-tioxypyrimidine and a dimer at pH = 3.0, and 5-guanidinohydantoin at pH = 7.0 [3, 27, 59]. The 8-oxodG electroanalytical determination at an edge plane surface of a PGE showed a linear range 5 × 10^{-7}–1 × 10^{-4} M and a LOD = 28 × 10^{-9} M [27].

Table 1 Electroanalytical determination of 8-oxoG and 8-oxodG at different NsM-modified electrode surfaces, from standard solutions of 8-oxoG or 8-oxodG

Target	Electrode	Method	Electrode modification	Linear range	LOD	Ref.
Bare electrodes						
8-oxoG	GCE	DPV	—	6×10^{-6}–2.8×10^{-5} M	0.8×10^{-6} M	[25]
	BDD	DPV	—	0–11.4×10^{-6} M	1.0×10^{-6} M	[26]
8-oxodG	PGE	SWV	—	5×10^{-7}–100×10^{-6} M	28×10^{-9} M	[27]
Carbon nanostructure–modified electrodes						
8-oxoG	GCE	LSV	SWCNTs	2.990×10^{-9}–8.725×10^{-5} M	1.0×10^{-9} M	[28]
	GCE	CV	MWCNTs	5.63×10^{-8}–1.64×10^{-5} M	0.2×10^{-9} M	[29]
	GCE	SWV	rGO-MWCNTs	3–75×10^{-6} M	35×10^{-9} M	[30]
	GCE	DPV	Sulphur-doped rGO	2×10^{-9}–20 M	1×10^{-9} M	[31]
	GCE	DPV	PtNF-GO	0.7×10^{-9}–2×10^{-6} M	25×10^{-6} M	[32]
	SPCE	DPV	MWCNTs-COOH	0.3–12 μmol dm^{-3}	0.57 μmol dm^{-3}	[33]
Carbon nanocomposite–modified electrodes						
8-oxodG	GCE	LSV	SWCNTs/L-lysine	3.0×10^{-7}–1.0×10^{-5} M	97×10^{-9} M	[34]
	GCE	DPASV, DPV	CNT/PEI	5.0×10^{-7}–3.0×10^{-5} M	0.1×10^{-6} M	[35]*
	GCE	DPASV	MWCNTs/Nafion	5×10^{-7}–2×10^{-5} M	13×10^{-9} M	[36]*
	GCE	DPASV	MWCNTs/percoll	5×10^{-7}–2×10^{-5} M	255×10^{-9} M	[36]*
	GCE	DPV	SWCNTs/Nafion	0.03–1.25×10^{-6} M	8.0×10^{-9} M	[37]
	GCE	LSV	Graphene/Nafion	0.07–33.04×10^{-6} M	1.12×10^{-9} M	[38]
	GCE	DPV	COOH-GO/MWCNTs/PEI/Au NPs	0.014–1.41×10^{-6} M	7.1×10^{-9} M	[39]
	GCE	DPV	Poly(L-arginine)/graphene/Au NPs	1.0–100×10^{-9} M	1.0×10^{-9} M	[40]
	GCE	DPV	GN/HKUST-1 NPs	10×10^{-9}–1×10^{-6} M	2.5×10^{-9} M	[41]
	GCE	DPV	Chitosan/zirconia	10–200 ng mL^{-1}	3.67 ng mL^{-1}	[42]
	GCE	Amperometry	Cu_2O/g-C_3N_4	25×10^{-9}–0.91×10^{-3} M	4.5×10^{-9} M	[43]
	GCE	DPV	rGO/ZnO NPs	5.0–5000.0×10^{-9} M	1.3×10^{-9} M	[44]
	SPCE	Amperometry	rGO/Dy_2O_3 NPs	0.05–135.3×10^{-6} M	1.0×10^{-9} M	[45]
	GCE	CV	ssDNA/GNs	0.0056–1.155×10^{-6} M	0.9×10^{-9} M	[46]
	—	Stochastic	rGO	1.0×10^{-14}–1.0×10^{-12} M	1.0×10^{-14}	[47]
	—	Stochastic	TiO_2-Au/rGO	1.0×10^{-14}–1.0×10^{-12} M	1.0×10^{-14}	[47]
	—	Stochastic	TiO_2-Ag/rGO	1.0×10^{-14}–1.0×10^{-12} M	1.0×10^{-14}	[47]
	Paper-based carbon	Chronoamperometry	CNs/immunochromatographic strips	0–150 ng mL^{-1}	3.11 ng mL^{-1}	[48]
	Paper-based carbon ink	DPV	PEDOT NPs	50–1000 ng mL^{-1}	14.4 ng mL^{-1}	[49]
Polymeric film–modified electrodes						
8-oxodG	GCE	CV	P3MT	0.700–70.0×10^{-9} M	0.1×10^{-6} M	[50]
	GCE	CV	DNA–P3MT	0.28–19.6×10^{-6} M	56×10^{-9} M	[51]
	GCE	DPV	P3AT	0.5–35×10^{-3} M	31×10^{-9} M	[52]
	PGE	SWV	MIP	20×10^{-9}–3×10^{-6} M	3×10^{-9} M	[53]
	Au	EIS	MIP	0.1–100 pg mL^{-1}.	0.74 pg mL^{-1}	[54]
Dendrimer-modified electrodes						
8-oxodG	Au	HPLC-AD	AET/$DG^{3.5}$ or $DG^{4.5}$ PAMAM dendrimers	—	1.2×10^{-9} M	[55]
Antibody-modified electrodes						
8-oxodG	GCE	DPV	PICA/chitosan/PA/anti-8-oxodG	0.1–10,000 $ngmL^{-1}$	30 $pgmL^{-1}$	[56]
	Au	EIS	amine/anti-8-oxodG	0.02–7.0 ng mL^{-1}	—	[57]
	Au	CV	ZnO NRs/anti-8-oxodG	0.001–5.00 ng mL^{-1}	100 fg mL^{-1}	[58]

*In the presence of ascorbic acid interference



Scheme 4 Schematic representation of NsM electrode surface modifiers

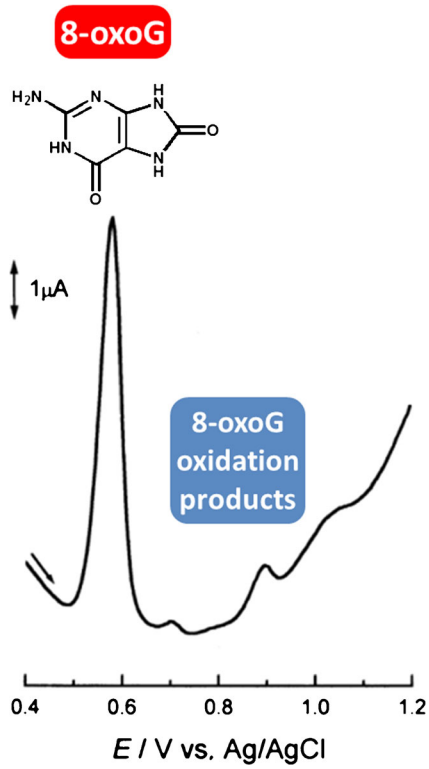


Fig. 1 DPV at a GCE of 0.10 mM 8-oxoG, in pH = 3.6. Adapted from [25] with permission

The 8-oxoG direct electroanalytical determination in urine samples poses a number of challenging problems, due to the presence of uric acid (UA), whose chemical structure is similar to 8-oxoG (Fig. 3) and is excreted in urine at concentrations of ~ 0.5 mM (25–70 mg L^{-1}), 10^4 times higher than the 8-oxoG urine levels. Similar to 8-oxoG, UA is electroactive, and follows a reversible pH-dependent oxidation mechanism, with the transfer of two electrons and two protons. The 8-oxoG oxidation in the presence of UA, at GCE, by DPV, in the pH range from 1 to 12, was studied [60]. The difference between the anodic peak potentials of 8-oxoG and UA was found to be less than 100 mV over the whole pH range. The pH = 6.0 was found to be optimum for the 8-oxoG electrochemical determination in the presence of UA, in mixtures of both analytes (Fig. 3) [60].

The 8-oxodG oxidation mechanism is similar at different electrode materials, such as PGE, GCE, platinum, gold, and Tin (IV) oxide electrodes [3, 27, 59]; however, the rate of the charge transfer reaction step depends on the electrode material, following the sequence GCE > Pt and Au >> SnO_2 [59].

Recently, different NsM-based surface electrode modification procedures, mainly using a GCE surface, have been proposed, greatly increasing the surface area of the electrode and improving the sensitivity and detection limits of the electroanalytical procedure for the quantification of the 8-oxoG and 8-oxodG oxidative DNA damage biomarkers.

Carbon NsM–modified electrodes

Electrochemical sensors based on carbon NsM–modified electrodes can improve the sensitivity and selectivity for the 8-oxoG and 8-oxodG electroanalytical detection, also presenting the advantages of low-cost and easy miniaturization, automation, and operation. Carbon NsM, such as CNT, graphene, buckypaper, and nanohybrids, have received much attention due to their high surface-to-volume ratio, high electrical conductivity, chemical stability, biocompatibility, and robust mechanical strength.

In particular, CNT present outstanding electrical, chemical, mechanical, and structural properties, enhancing the electrochemical properties of chemical compounds and biomolecules, and promoting their electron-transfer reactions [61]. In terms of structure, there are two groups of CNT, single-wall carbon-nanotubes (SWCNT) that have a cylindrical nanostructure, formed by rolling up a single graphite sheet into a tube, and multi-wall carbon nanotubes (MWCNT) that are formed by several layers of concentrically graphene cylinders (Scheme 4).

A porous SWCNT–modified GCE was used to develop a sensor for the electroanalytical detection of 8-oxodG,

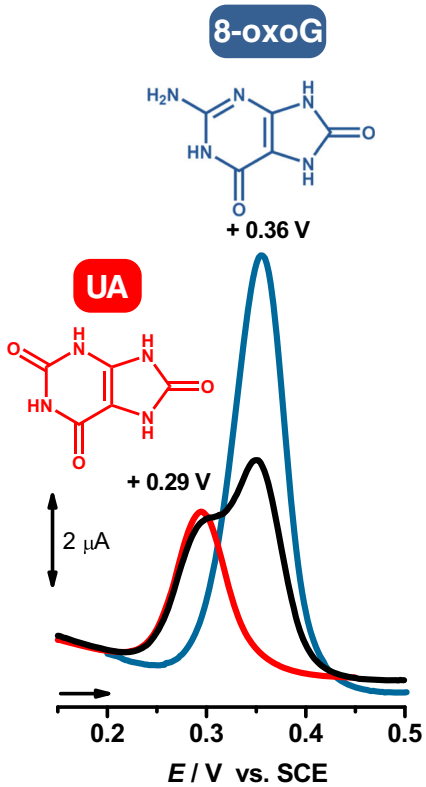


Fig. 3 DPVs at a GCE: (blue line) 5 μM 8-oxoG, (red line) 5 μM UA, and (black line) 5 μM equimolar mixture of 8-oxoG and UA, in pH = 6.2. Adapted from [60] with permission

presenting two linear ranges 2.990×10^{-9}–3.061×10^{-6} M and 3.061×10^{-6}–8.725×10^{-5} M, with a LOD = 1.0×10^{-9} M [28], and showed promising results for the simultaneous determination of 8-oxodG, UA, and dopamine. A sensor for the 8-oxodG electroanalytical determination, based on

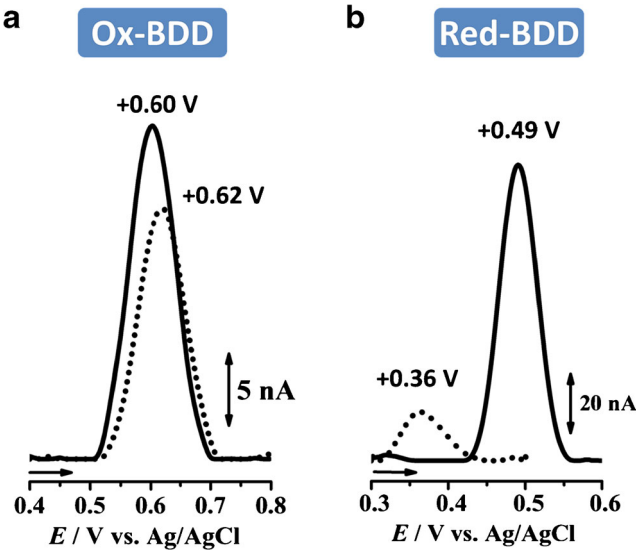


Fig. 2 DPV baseline-corrected of 8-oxoG in (black line) pH = 4.5 and (dotted line) pH = 7.0, at a BDDE, after: **a** anodic (Ox-BDDE) and **b** cathodic (Red-BDDE) electrochemical pre-treatment. Adapted from [26] with permission

a MWCNTs modified GCE, showed a good electrochemical response, also in two linear ranges 5.63×10^{-8}–6.08×10^{-6} M and 6.08–16.4×10^{-6} M, with a LOD = 1.88×10^{-8} M [29].

Graphene is another carbon NsM of choice for electrochemical sensor applications [62]. Among different forms of graphene, reduced graphene oxide (rGO) is manufactured by the oxidation/exfoliation of graphite to graphene oxide (GO), followed by its chemical, thermal, or electrochemical reduction. Generally, rGO is preferred for the fabrication of electrochemical sensor devices, due to the large number of functional groups at its surface, such as hydroxyl, epoxides, and –COOH, that gives rGO the ability to interact and bind easily with different chemical compounds.

A voltammetric sensor for the determination of 8-oxodG, based on the GCE surface modified by rGO and MWCNTs, showed a sensitivity of 0.1965 μA μM^{-1}, in the linear range 3–75×10^{-6} M, with a LOD = 35×10^{-9} M [30]. This sensor was successfully applied for the determination of 8-oxodG, in the presence of ascorbic acid (AA), UA, xanthine, and hypoxanthine interferences and in human urine samples.

A sulphur-doped rGO–modified GCE was used to develop an electrochemical sensor for the detection of 8-oxodG in the linear range 2×10^{-9}–20 M, with a LOD = 1×10^{-9} M [31].

A controllable flower-like Pt-graphene oxide (PtNF-GO) architecture was synthesized through a layer-by-layer electrostatic self-assembly and was immobilized onto a GCE [32]. The developed sensor was used for the electroanalytical quantification of 8-oxodG, in two linear ranges 0.7×10^{-9}–2×10^{-6} M and 2–22×10^{-6} M, with a low LOD = 0.025×10^{-9} M.

The electrochemical detection of 8-oxoG at a disposable screen-printed carbon electrode (SPCE), unmodified and modified with carboxy-functionalized MWCNTs, was also investigated [33]. The MWCNTs-COOH modification increased the surface area, thus also increasing the current and sensor sensitivity, 8-oxoG being determined in the linear range 0.3–12×10^{-6} M, with the LOD = 0.57×10^{-6} M.

Carbon nanocomposite–modified electrodes

Aiming to overcome problems related with the use of single nanomaterials, e.g. CNT adhesion, curling, and stacking of graphene lamellae, difficulties in achieving dispersion of metal NPs, single NsM started to be functionalized and/or combined with other inorganic and organic functional materials to form nanocomposites.

The SWCNTs covalently functionalized with L-lysine, immobilized at the surface of GCE, have been used for the sensitive electrochemical quantification of 8-oxodG, G, and adenine, by voltammetric adsorptive stripping with medium exchange [34]. The sensor achieved the 8-oxodG detection in the linear range 3–100×10^{-7} M, with a LOD = 97×10^{-9} M.

Polymer nanocomposites, formed by combining carbon NsM into a polymer matrix, showed enhanced mechanical strength and high electrical properties. The GCE surface modified by CNT dispersed in polyethylenimine (PEI) have been used for the 8-oxodG electrochemical detection in the presence of AA and UA, in the linear range 5–300 × 10^{-7} M, with a LOD = 1.0 × 10^{-7} M and a LOQ = 3.0 × 10^{-7} M [35].

An electrochemical sensor device for the detection of 8-oxodG and UA in the presence of AA has also been developed at a GCE modified by MWCNTs dispersed in various polymers such as Nafion, polyacrylic acid, percoll, and polyvinyl alcohol [36]. The 8-oxodG determination in the presence of AA, in the linear range 5–200 × 10^{-7} M, presented the best results at the MWCNTs/Nafion modified GCE, with a LOD ~ 13 × 10^{-9} M, when compared with MWCNTs/percoll modified GCE, with a LOD = 2.6 × 10^{-7} M.

An electrochemical sensor, based on a GCE modified by a SWCNTs/Nafion nanocomposite film, was also developed [37]. Due to the Nafion strong cation-exchange ability and the SWCNT electronic properties, the sensor showed a good response for 8-oxodG in the linear range 0.03–1.25 × 10^{-6} M, with a LOD = 8.0 × 10^{-9} M. An electrochemical sensor, based on a graphene/Nafion nanocomposite film immobilized at a GCE, was also reported [38], and used for the electroanalytical determination of 8-oxodG, in the linear range 0.07–33.04 × 10^{-6} M, with a LOD = 1.12 × 10^{-9} M, the Nafion helping to reduce the UA anodic current.

Based on a GCE modified by carboxyl-functionalized graphene oxide (GO), carboxyl-functionalized MWCNTs, and PEI-wrapped Au NPs, a sensor for the detection of urinary 8-oxodG was developed (Scheme 5) [39], showing a good response in tree linear ranges 0.014–0.14 × 10^{-6} M, 0.14–1.41 × 10^{-6} M, and 1.41–14.12 × 10^{-6} M, with a LOD = 7.06 × 10^{-9} M. The UA interference in urine samples was reduced by uricase pre-treatment.

A GCE modified by a film of poly(L-arginine) and graphene-wrapped Au NPs, for the DPV quantification of 8-oxodG, in the presence of interfering UA and AA, in the linear ranges 1.0–100 × 10^{-9} M and 0.5–10 × 10^{-6} M, with a LOD = 1.0 × 10^{-9} M, have been used [40].

An electrochemical sensing platform for the detection of 8-oxodG was developed using ultrasmall Cu-based metal organic frame (HKUST-1) NPs anchored on the surface of graphite nanosheets (GN), further immobilized at the surface of GCE [41]. The sensor presented a wide linear range 10 × 10^{-9}–1 × 10^{-6} M and a LOD ~ 2.5 × 10^{-9} M.

More recently, an electrochemical sensor based on a chitosan/zirconia composite modified GCE, was developed [42]. The sensor achieved, by DPV, the 8-oxodG detection in the linear range 10–200 ng mL^{-1}, with a LOD = 3.67 ng mL^{-1}, and presented also good stability and sensitivity in urine samples.

An electrochemical sensor, based on GCE modification by cuprous oxide (Cu_2O) cubes decorated with graphitic carbon nitrides (g-C_3N_4), was reported [43]. The 8-oxodG accumulated on the Cu_2O/g-C_3N_4-modified electrode enhanced the amperometric response, and a wide linear range, 25 × 10^{-9}–0.91 × 10^{-3} M, and a LOD = 4.5 × 10^{-9} M for 8-oxodG were achieved. The sensor showed good stability, repeatability, and reproducibility after a long storage time at 4 °C.

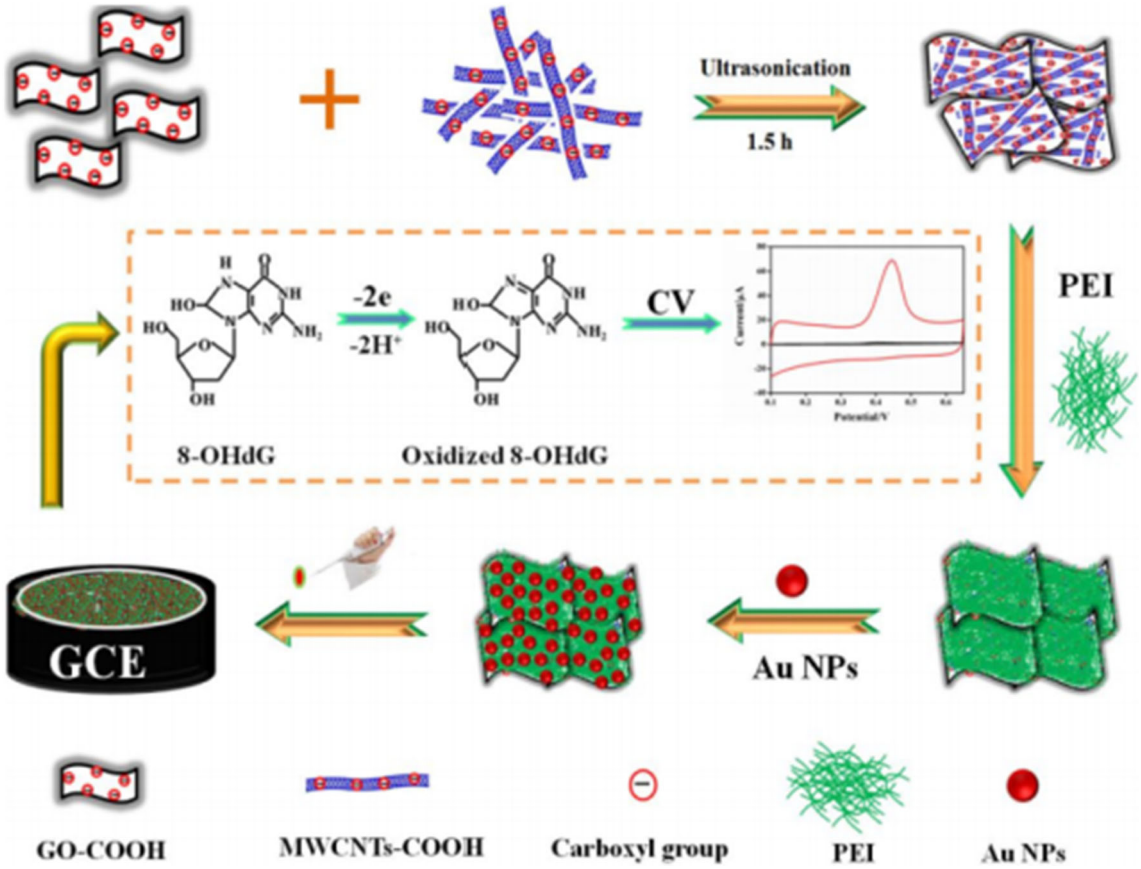
A GCE modified by a nanocomposite of rGO with ZnO NPs was used for the determination of 8-oxodG, by DPV, in the linear range 5.0–5000.0 × 10^{-9} M, with a LOD = 1.25 × 10^{-9} M [44]. The ZnO NP incorporation enhanced the 8-oxodG oxidation signal when compared with bare GCE and rGO-modified GCE. The developed sensor showed the possibility of application in urine samples, after using uricase to efficiently eliminate the UA interference.

The rGO embedded with dysprosium oxide (Dy_2O_3) NPs immobilized at the surface of SPCE, for the development of an electrochemical sensor for the amperometric detection of 8-oxodG in human blood and urine samples, was used [45]. The amperometric sensor was able to detect 8-oxodG in the linear range 0.05–135.3 × 10^{-6} M, with a LOD = 1.02 × 10^{-9} M. The method was successfully tested in human urine and blood serum samples, with results comparable with those obtained by HPLC.

A 8-oxodG biosensor, consisting on a GCE surface modified by single-stranded DNA (ssDNA) and functionalized graphene nanosheets (GNs) nanocomposites, was developed [46], which showed a good electrocatalytic activity towards 8-oxodG oxidation, with a linear range 0.0056–1.155 × 10^{-6} M and a LOD = 0.875 × 10^{-9} M.

The electroanalytical determination of 8-oxodG and 8-nitroguanine in blood and urine samples was successfully achieved using three stochastic sensors, based on different rGO powders: reduced nanographene oxide (nGO), rGO with embedded TiO_2 and Ag NPs (TiO_2-Ag/rGO), and rGO with embedded TiO_2 and Au NPs (TiO_2-Au/rGO); all three composites were also modified by a protoporphyrin IX–zinc complex [47]. The TiO_2, Ag, and Au incorporation increased the graphene oxide conductivity, allowing the 8-oxodG analytical determination in the liner range 1–100 × 10^{-14} M, with a LOD = 1.0 × 10^{-14} M, for all these sensor devices. However, the highest sensitivity for the 8-oxodG and 8-nitroguanine detection was obtained using TiO_2-Au/rGO composites, suggesting their possible application for quantitative determinations for leukaemia and gastric cancer diagnosis.

A sensor device for the quantitative analysis of 8-oxodG, which integrated paper electrodes, CNTs and immunochromatographic strips, using a combined electrochemical and colorimetric detection, was reported [48]. In the linear range 0–150 ng mL^{-1}, the electrochemical sensor achieved a LOD = 3.11 ng mL^{-1} in standard solutions and a LOD = 8.85 ng mL^{-1} in urine samples.



Scheme 5 Schematic representation of the 8-oxodG electrochemical detection and preparation process of the GCE modified by COOH-GO/ MWCNTs/ PEI/Au NP nanocomposites. Reproduced from [39] with permission

Different label-free sensors for the 8-oxodG detection, which used a paper-based carbon-ink electrode, modified by poly(3,4-ethylenedioxythiophene) (PEDOT) NPs, MWCNT, or MWCNT-COOH powders, have been developed [49]. The sensor incorporating PEDOT showed the best selectivity, reproducibility, stability, and sensitivity, in the linear range 50–1000 ng mL^{-1}, with a LOD = 14.4 ng mL^{-1}.

Polymeric film–modified electrodes

Conducting polymers present unique electrical properties and have emerged as promising NsM in electrochemical sensor applications.

The 8-oxodG electroanalytical determination, at an electropolymerized poly(3-methylthiophene) (P3MT) modified GCE, using CV and linear sweep voltammetry, was performed [50]. At the P3MT-modified GCE, the 8-oxodG oxidation peak shifted to more negative potentials, presenting enhanced oxidation peak current, when compared with bare GCE. The sensor achieved the 8-oxodG detection, in the linear ranges $0.70–35.0 \times 10^{-6}$ M and $35.0–70.0 \times 10^{-9}$ M, with a LOD = 0.10×10^{-6} M [50]. In another report, a mixed DNA–P3MT film immobilized at the GCE surface was described [51], which showed the detection of 8-oxodG, in the linear ranges $0.28–4.2 \times 10^{-6}$ M and $4.2–19.6 \times 10^{-6}$ M, with a LOD = 56×10^{-9} M.

Using an electrochemical sensor based on a GCE modified by electropolymerized poly(3-acetylthiophene) (P3AT) film, the 3-chloro-1,2-propanediol-induced cytotoxicity and 8-oxodG generation, via ROS-mediated oxidative DNA damage in HEK293 cells, was detected [52]. The sensor was used for the electroanalytical detection of 8-oxodG in the oxidation-damaged cells, showing the linear range $0.5–35 \times 10^{-3}$ M and a LOD = 31.3×10^{-9} M.

The $[Os(bpy)_2(PVP)_{10}Cl]^+$ and $[Ru(bpy)_2\text{-}(PVP)_{10}Cl]^+$ metallopolymers were assembled layer by layer on PGE, in order to develop a sensor selective for 8-oxoG and G [63].

Molecularly imprinted polymers (MIPs) are synthetic polymers that present a predetermined selectivity for a given analyte. In the MIP fabrication process, the target analyte is present during the formation of the three-dimensional structure of the polymer. The subsequent removal of the target analyte template leaves binding sites within the polymer with the shape and orientation of the functional groups corresponding specifically to the target analyte, which are subsequently used for specific recognition, by rebinding the analyte, in electrochemical sensor applications.

A MIP-based sensor for the determination of 8-oxodG has been fabricated by electropolymerization of melamine in the presence of 8-oxodG, on glutaraldehyde/poly-1,5-diaminonaphthalene–modified edge plane PGE [53]. The sensor response was in the linear range 20×10^{-9}–3×10^{-6} M and presented a LOD = 3×10^{-9} M, showing no interference from UA, AA, xanthine, and hypoxanthine up to 100-fold concentration.

A promising impedimetric sensor for the detection of urinary 8-oxodG, based on a MIP assembled at the surface of Au electrodes, has been reported [54]. The sensor was able to rebind 8-oxodG, in the linear range 0.1–100 pg mL^{-1} (3.5–3500×10^{-12} M), with a LOD = 0.74 pg mL^{-1}, detected by electrochemical impedance spectroscopy (EIS).

Dendrimer-modified electrodes

Dendrimers are a synthetic class of highly branched spherical polymers, formed by a central core, multiple peripheral reactive site end groups, and branching repeating units. In particular, poly(amidoamine) (PAMAM) dendrimer terminal functional groups –COOH and $–NH_2$ showed the ability to promote the 8-oxodG interfacial concentration [55].

An electrochemical sensor surface, based on an Au electrode thiol-modified with self-assembled monolayers of different generation of PAMAM dendrimers, was used for 8-oxodG determination by HPLC-ECD [55]. The best results were achieved at Au electrodes modified with aliphatic amino thiols (AET) and 3.5 and 4.5 generation carboxylated dendrimers ($AET/DG^{3.5}$ and $AET/DG^{4.5}$). This mixed methodology, electrochemical sensor detection after HPLC separation, allowed the 8-oxodG detection in synthetic samples with a LOD = 1.2×10^{-9} M and a LOQ = 3.7×10^{-9} M, avoiding sample pre-treatment [55].

Antibody-modified electrodes

An electrochemical immunosensor is a biosensor device employing the antibody as a capture agent, which quantitatively measures the electrochemical signal resulting from the binding event between the antibody and the analyte target molecule. Based on the antigen-antibody very specific recognition, the electrochemical immunosensors have received much attention, due to their excellent selectivity and LOD, for a small sample volume. However, the antigen-antibody interact through a high affinity binding, much like lock and key, and no electroactive products are formed. Therefore, the immunosensors assembly generally have complicated modifications, with derivatization and employing different NsM, in order to enable the electrochemical detection.

An immunosensor for the detection of 8-oxodG, based on GCE modification by poly(indole-5-carboxylic acid) (PICA), chitosan, dropping protein A (PA), and 8-oxodG antibodies (anti-8-oxodG) as biorecognition element, was developed [56]. The PICA/chitosan modification increased the electrode surface area for the immunoresponse, promoting the electron transfer kinetics. The immunosensor showed the linear range $0.1–1.0 \times 10^4$ ng mL^{-1} and a LOD = 30 pg mL^{-1}.

The use of 8-oxodG antibodies attached to phenyl amine functionalized graphene surfaces led to the development of an immunosensor that allowed the 8-oxodG detection in the linear range 0.1–20 ng mL^{-1}, with a LOD = 0.1 ng mL^{-1} (0.35×10^{-9} M) [64].

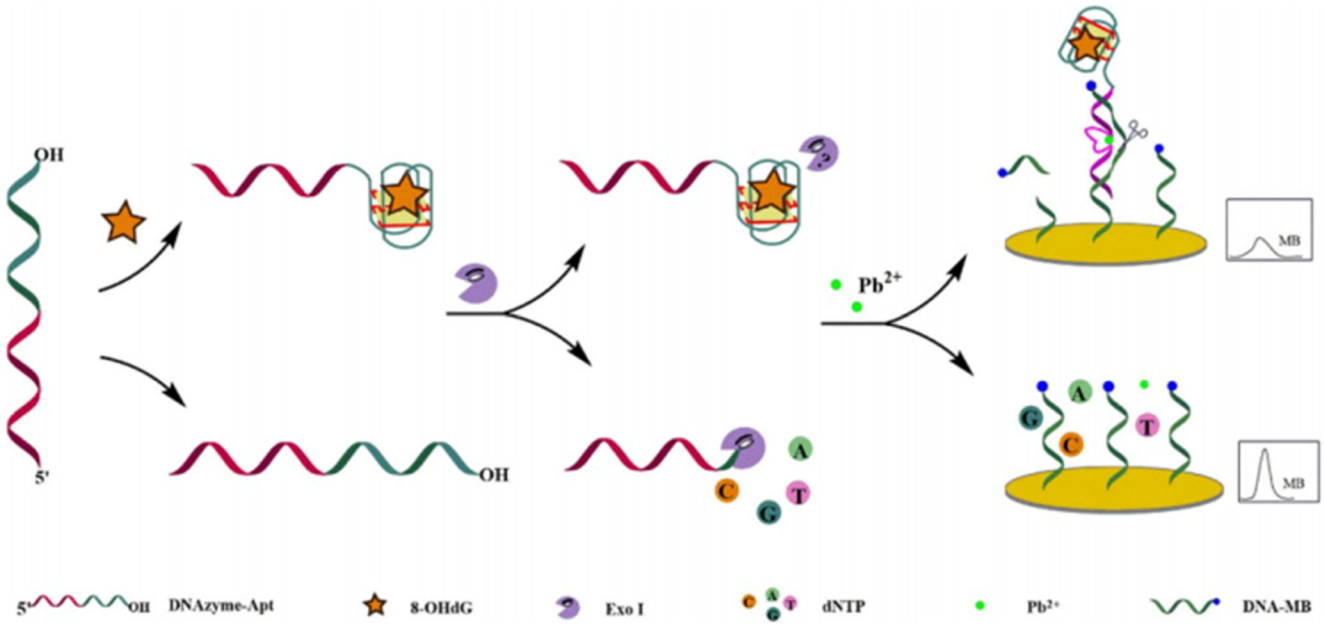
An impedimetric immunosensor, consisting on an amine-modified Au electrode as a platform for the 8-oxodG antibodies immobilization, was prepared [57]. The 8-oxodG electroanalytical determination, in the linear range 0.02–7.0 ng mL^{-1}, was followed by EIS, and glucose, urea, and creatinine interferences were found insignificant.

An immunosensor for the detection of 8-oxodG, which used a printed circuit board (PCB) technology, to prepare Au film–based electrodes, modified either with ZnO nanorods (NRs) or rGO/ZnO NR composites, as a platform for the 8-oxodG antibodies immobilization, was developed [58]. The ZnO NRs/anti-8-oxodG immunosensor presented a wider linear range 0.001–5.00 ng mL^{-1} and a LOD = 100 fg mL^{-1}, being suitable for the 8-oxodG detection in urine samples and showed better reproducibility than the rGO/ZnO NRs/anti-8-oxodG immunosensor [58].

Aptamer-modified electrodes

Aptamers are nucleic acid or peptide sequences selected from a large random sequence pool, for their ability to bind specifically to a target analyte. In particular, nucleic acid aptamers are small, highly structured DNA/RNA sequences, isolated from combinatorial libraries, in a chemical synthesis, by SELEX (systematic evolution of ligands by exponential enrichment). They present remarkable molecular recognition properties, by modifying their conformation after binding the target analyte, and started to be recently used for the detection of 8-oxodG, due to their better stability, lower size, lower associated costs, and without using animals, when compared with protein antibodies.

The electroanalytical determination of 8-oxodG, based on nucleic acid sequences containing an aptamer and a DNAzyme (Scheme 6) was achieved [65]. The Au electrode surface modified by DNA sequences labelled with methylene blue (MB) redox probes (the MB-DNA sequences) was immersed in solutions containing the 8-oxodG target, aptamer-DNAzyme sequences, exonuclease Exo I enzyme, and Pb^{2+} ions, and EIS and SWV experiments were performed. In the solution, the aptamer part of the aptamer-DNAzyme sequence specifically recognizes and bounds the 8-oxodG, which induces the aptamer conformational modification from a single-strand into a G-quadruplex, which further protects the



Scheme 6 Schematic representation of the 8-oxodG electrochemical detection, based on exonuclease-mediated functional nucleic acid. Reproduced from [65] with permission

aptamer-DNAzyme sequence from degradation by the Exo I. The aptamer-DNAzyme sequences that are not bound to the 8-oxodG are degraded by Exo I, suppressing the background noise and enhancing the sensitivity of the EIS assay. Then, in the presence of Pb^{2+} ions, the DNAzyme part of the aptamer-DNAzyme sequence forms a deoxyribozyme structure with the MB-DNA sequence immobilized at the Au electrode, releasing the MB probe and decreasing the electrochemical current. Using this approach, 8-oxodG was detected, in the linear range 0.01×10^{-9}–7.0×10^{-6} M, with the LOD = 6.82×10^{-12} M, this method being also successfully applied for the determination of 8-oxodG in urine [65]. The 8-oxodG detection specificity in the presence of UA, AA, citric acid, glucose, and metal ions, such as Cu^{+2}, Zn^{2+}, Mg^{2+}, and Al^{3+}, was also investigated.

An enzyme-free electrochemical aptasensor for the determination of 8-oxodG, based on the hybridization chain reaction (HCR) signal amplification, was also reported [66]. A capture DNA was first immobilized onto an Au electrode and used to hybridize one terminus of the 8-oxodG aptamer, while the other aptamer terminus hybridized an auxiliary DNA sequence. The auxiliary DNA sequence was further used as an initiator to propagate the HCR and led to the formation of extended dsDNA structures on the electrode surface. However, in the presence of 8-oxodG, the aptamer changed its conformation from a single strand into a G-quadruplex, which made impossible the HCR, leading to a signal decrease. The 8-oxodG presence was detected by DPV, via a $[Ru(NH_3)_6]^{3+}$ redox label, in the linear range 10×10^{-12}–100×10^{-6} M, with the LOD = 2.5×10^{-12} M [66].

In another report, an enzyme-free electrochemical aptasensor for the detection of 8-oxodG, which employed three-signal amplification strategies to enhance the sensitivity, was developed [67]. The developed 8-oxodG detection method started with solutions of the 8-oxodG target incubated with aptamer/pDNA complexes immobilized on Fe_3O_4 magnetic beads. In the presence of 8-oxodG, the aptamer/pDNA complexes could release three types of pDNA sequences in the presence of 8-oxodG target, which caused the first 3-fold magnification of the 8-oxodG target. Second, the released three types of pDNAs initiated catalysed hairpin assembly between two hairpin DNAs (HP1 and HP2) immobilized on the surface of an Au electrode. The HP2 further induced a HCR between two hairpin DNAs (HP3 and HP4), forming a long double-stranded DNA concatemer on the surface of the Au electrode. The electrochemical detection was achieved by DPV, using a $[Ru(NH_3)_6]^{3+}$ redox label absorbed onto the dsDNA concatemer by electrostatic interactions. The strategy led to a 8-oxodG electrochemical detection in the linear range 100×10^{-15}–10×10^{-9} M, with a LOD = 24.34×10^{-15} M [67].

Conclusions

The predominant forms of free radical-induced oxidative DNA damage, in nuclear and mitochondrial DNA, are 8-oxoG and its corresponding nucleotide 8-oxodG, and they have been used as biomarkers for risk assessment of oxidative stress, cancer, degenerative, and other age-related diseases.

Both biomarkers 8-oxoG and 8-oxodG undergo a reversible, pH-dependent oxidation process at carbon electrodes,

which occurs at the purine moiety, with the transfer of two electrons and two protons, followed by the 8-oxoG and 8-oxodG electroactive oxidation products' irreversible reactions.

Electroanalytical methods are an extremely advantageous methodology to evaluate the levels of oxidative DNA damage biomarkers 8-oxoG and 8-oxodG in urine, blood, and tissue samples, due to their low detection limit, simplicity, celerity, and economic consumption of samples, and NsM-modified electrodes have recently gained an increased attention. This review summarizes the fundamental principles concerning the use of NsM-based electrochemical sensors for the 8-oxoG and 8-oxodG electroanalytical determination, via electrode surface modification by carbon nanostructures, carbon nanocomposites, polymeric films, dendrimers, antibodies, and aptamers.

The development of NsM-based electrochemical sensors for detecting 8-oxoG and 8-oxodG biomarkers of oxidative damage in biological fluids clearly shows very exciting perspectives. Nevertheless, also face different challenges, such as lower LOD of the target in biological fluids, the presence of interferences, fouling of the sensor surface due to nonspecific adsorption of the proteins, other biological components, and/or pharmaceutical drugs, and low long-term stability of the sensing device in complex matrixes. Therefore, complex samples of urine, blood plasma, and tissue culture medium may require preliminary steps of separation, prior to the 8-oxoG and 8-oxodG electrochemical identification and quantification, and future trends in electroanalytical determination of 8-oxoG and 8-oxodG biomarkers should combine the HPLC efficiency in separation with the advantages of electrochemical detection at NsM-based sensors.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Abbreviations AET/DG$^{3.5}$ and AET/DG$^{4.5}$, 3.5 and 4.5 generation carboxylated dendrimers; BDDE, 8-oxodG antibodies (anti-8-oxodG), boron-doped diamond electrode; CNT, carbon nanotubes; Cu_2O, cuprous oxide; CV, cyclic voltammetry; DPASV, differential pulse anodic stripping voltammetry; DPV, differential pulse voltammetry; PA, dropping protein A; Dy_2O_3, dysprosium oxide; PtNF-GO, flower-like Pt-graphene oxide; GCE, glassy carbon electrode; GNs, graphene nanosheets; GO, graphene oxide; GN, graphite nanosheets; g-C_3N_4, graphitic carbon nitrides; HPLC-AD, high-performance liquid chromatography with amperometric detection; LSV, linear sweep voltammetry; MIP, molecularly imprinted polymer; MWCNTs, multi-walled carbon nanotubes; NPs, nanoparticles; NRs, nanorods; PEDOT, poly(3,4-ethylenedioxythiophene); P3AT, poly(3-acetylthiophene); P3MT, poly(3-methylthiophene); PAMAM, poly(amidoamine); PICA, poly(indole-5-carboxylic acid); PEI, polyethyleneimine; rGO, reduced graphene oxide; TiO_2-Ag/rGO, rGO with embedded TiO_2 and Ag NPs; TiO_2-Au/rGO, rGO with embedded TiO_2 and Au NPs; SPCE, screen-printed carbon electrode; ssDNA, single-stranded DNA; SWCNTs, single-walled carbon nanotubes; SWV, square wave voltammetry; MWCNT, multi-walled carbon nanotubes

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